

HAT-P-12b

R-0710

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-12_250511 · created 2026-07-27T22:43:32+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460806.7497 +/- 0.0047 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.14 +/- 0.05
Transit depth (Rp/Rs)^2	1.96 +/- 1.41 [%]
Orbital Inclination (inc)	89.98 +/- 1.85
Airmass coefficient 1 (a1)	1.09 +/- 0.028
Airmass coefficient 2 (a2)	-0.08 +/- 0.16
Scatter in the residuals of the lightcurve fit is	2.59 %
Best Comparison Star	#1 - [413, 117]
Optimal Method	PSF photometry
Transit Duration (day)	0.094 +/- 0.01

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.14 ± 0.05 vs 0.1406 (deviation 0.01σ)
Duration fit vs published	0.094 d vs 0.0975 d (deviation 0.25σ)
Mid-time O–C	-7.9 min
Depth SNR	2.0
Residual scatter	2.59 %

Light curve

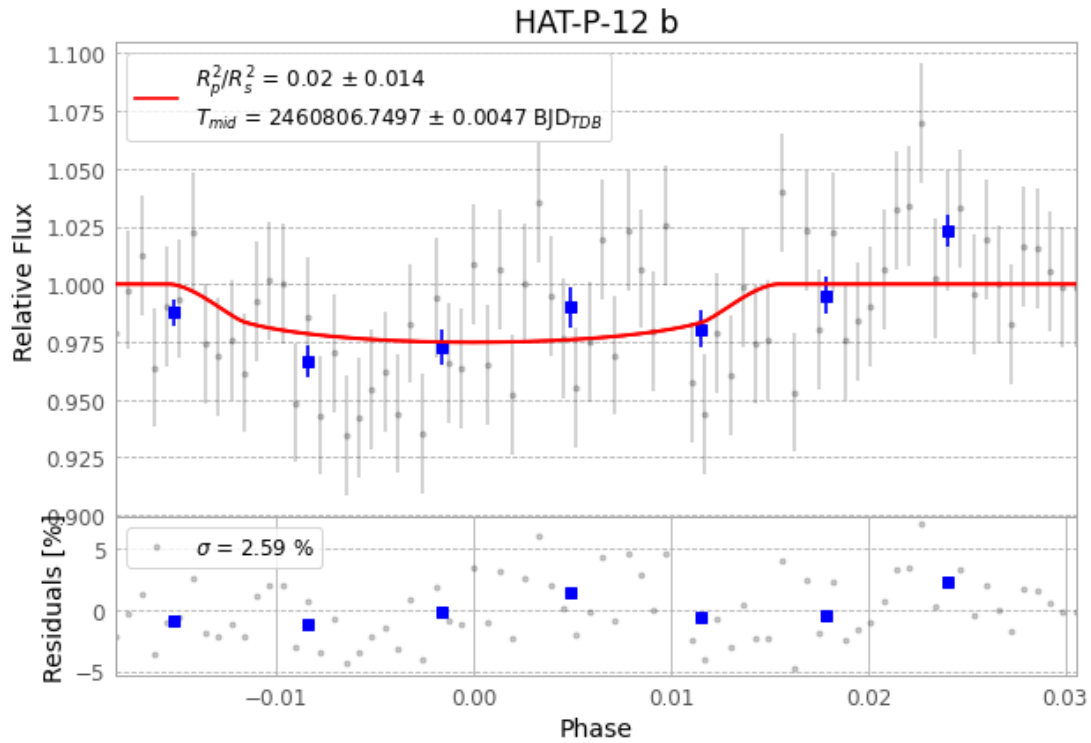


Figure 1 — FinalLightCurve_HAT-P-12 b_2025-05-10.png (SHA-256 ac6846dd6ba71097...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [344, 227], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 df7a880b5a9d2c71...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit f7446be

Data provenance

76 FITS frames (50 MB), HATP-12250511043015.FITS → HATP-12250511081515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-12-250511.log (68a0811c40cd...), FinalLightCurve_HAT-P-12 b_2025-05-10.png (ac6846dd6ba7...), FinalLightCurve_HAT-P-12 b_2025-05-10.csv (7e1794d05a32...)

Compute resources

Wall time 140.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 22:43Z.